Prabhavit Gupta



Senior Backend Systems Engineer: Distributed Pipelines, APIs & Cloud Architecture

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Summary

Senior Backend Systems Engineer with 3 years of commercial production experience designing, building, and operating high-throughput distributed backends, streaming data pipelines, and mission-critical cloud APIs. Proven track record architecting event-driven microservices that process millions of events daily with 99.8% production reliability. Deep practical expertise across Python (FastAPI, asyncio), Node.js/TypeScript, PostgreSQL optimization, Redis message brokers, and containerized cloud architectures. Based in Venice, Italy (CET time zone) with an EU Long-Term Residence Permit (permanent unrestricted EU work authorization across all EU member states, no sponsorship required); available immediately for full-time remote roles.

Technical Expertise

- **Distributed Backend Systems:** High-throughput REST and asynchronous RPC APIs, event-driven microservice architectures, distributed task queues, rate-limiting, circuit breakers, and fault-tolerant service orchestration.
- **Data Pipelines & Streaming:** Python (FastAPI, asyncio, pandas, NumPy, Pydantic), Node.js, real-time telemetry ingestion, ETL data processing, third-party messaging integrations, and webhook consumers.
- **Databases & Caching:** PostgreSQL (advanced schema design, query optimization, indexing, ACID transactional boundaries), Redis (in-memory caching, pub/sub, atomic locking, queue management).
- **DevOps & Infrastructure:** Docker containerization, cloud service architecture (AWS / GCP), GitHub Actions CI/CD automation, Prometheus/Grafana metric telemetry, structured JSON logging.
- **Applied AI & Tooling:** Claude Code, Model Context Protocol (MCP) integrations, automated testing pipelines, and agentic workflows for accelerated system delivery.
- **Languages & Authorization:** English (C1 Professional, fluent), Hindi (Native), Italian (Conversational). EU Long-Term Residence Permit (unrestricted EU work authorization across all EU member states, no sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Backend & AI Systems Engineer *London, UK (remote)*

- Architected and scaled high-throughput backend services powering an enterprise AI platform supporting 4,500+ active users across European organizations with 99.8% uptime.
- Engineered asynchronous Python (FastAPI) and Node.js microservices handling real-time telemetry, automated data extraction, and high-concurrency client requests.
- Designed resilient database architectures in PostgreSQL and implemented distributed caching layers in Redis, reducing p95 database query latencies by 42%.
- Integrated robust third-party APIs and webhook processors with idempotent delivery guarantees, automated retries, and comprehensive error auditing.
- Automated end-to-end integration test suites and multi-stage GitHub Actions CI/CD deployment pipelines, ensuring zero-downtime release rollouts.

Urlink **Apr 2025 - Jul 2025**
Backend Solutions Developer (Freelance) *Barcelona, Spain (remote)*

- Built scalable data ingestion microservices and RESTful API endpoints for external data synchronization across distributed enterprise platforms.
- Optimized PostgreSQL database queries and transactional boundaries, preventing data race conditions during high-volume ingestion cycles.
- Designed comprehensive API specifications and automated testing harnesses using Playwright and pytest to validate system contracts before client handover.

Independent Technical Consultant **Aug 2023 - Feb 2024**
Backend Systems & Automation Developer *Venice, Italy*

- Engineered custom backend APIs, data pipelines, and asynchronous automation scripts using Python and TypeScript for European SMB clients.

- Implemented event-driven webhook listeners and database synchronization jobs, eliminating manual operational data entry.
- Produced detailed technical architecture documentation, database schema diagrams, and deployment runbooks for client operations.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous foundational study in formal logic, epistemology, systems theory, and mathematical economic models.